Measuring Advertising Effectiveness via Granular and Aggregate Inference Ensemble Fusion

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Abstract

Marketers often measure advertising effectiveness using one of two approaches. Macro-level analysis uses aggregate data (typically realized via Marketing Mix Modeling or MMM) and Micro-level analysis uses individual level data (typically realized via Multi-Touch Attribution or MTA). Although both of these approaches can be used to measure the returns of the same campaigns, they frequently yield divergent Return on Investment (ROI) estimates, making it difficult to allocate budgets. We develop a model to explain the source of this divergence, and provide a solution that we test in simulation and using empirical data. Our model focuses on a policy-relevant estimand: the marginal revenue generated by an additional impression allocated according to a specific targeting policy. We demonstrate that under ideal conditions, estimates from the Macro and Micro approaches are asymptotically equivalent estimators. However in realistic environments affected by targeting heterogeneity and model misspecification, the two approaches diverge systematically. Macro models can suffer from aggregation bias, while Micro models suffer from high estimation variance and functional misspecification. To address this, we develop an ROI Ensemble Fusion Framework that treats the Macro and Micro estimates as two potentially biased and noisy measurements of the same underlying true ROI. That is, even if the researcher has the granular level data (and could run "only" an MTA), they should (and maybe surprisingly so) form an aggregated version of the same data and run an MMM, and then fuse the two together. Using a large-scale dataset and extensive simulation, we demonstrate that our fusion approach yields estimates that are more stable and robust to time-varying volatility than either method alone. Our findings provide a decision-making solution to managers who encounter competing ROI estimates from the same campaign.

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1 Introduction

Marketers use ad effectiveness estimates such as return on investment (ROI) to allocate advertising budgets across channels, platforms, and campaigns. In many cases, the same campaign is evaluated through multiple data environments, analytical pipelines, and organizational units, resulting in different ROI estimates for the same campaign. In particular, two commonly used approaches routinely deliver conflicting estimates. The first, which we call the *Macro* approach, uses aggregate time-series data about spend and outcomes and is typically implemented through Marketing Mix Modeling (MMM) [12, 8, 13**?**]. The second, which we call the *Micro* approach, uses granular exposure and conversion data to measure ad effectivness, and is typically implemented through user-level response models such as Multi-Touch Attribution (MTA) [15, 1, 3]. As these two approaches operate under different information sets, different levels of aggregation, and different modeling assumptions the same marketer may face materially different ROI estimates, creating a fundamental decision problem. Macro estimates often inform strategic budgeting because they are stable, available over long horizons, and aligned with aggregate business outcomes. Micro estimates often inform tactical optimization because they exploit user-level exposure histories and targeting information. When the two disagree, the decision-maker is left without a principled way to determine which estimate should guide allocation. The problem is particularly acute in large organizations, where different teams, agencies, or platform interfaces may produce different performance assessments from the same underlying campaign. Despite much progress on different techniques to measure advertising effectiveness using aggregate data [12, 8, 13, 17], individual-level data [15, 1, 6], or even experimental approaches [14, 9, 10], the question of how to interpret and reconcile conflicting Macro and Micro measurements remains unresolved.

Our paper formalizes the ad effectiveness measurement problem (Section 3), analyzes the source of diverging estimates (Section 4) and develops a method (Section 5) to combine competing estimates into one, more precise and accurate estimate, that resolves the decision-making challenge. First, we note that both Micro and Macro approaches attempt to estimate the same estimand. We focus on the *policy-path marginal ROI*: the marginal outcome from one additional impression delivered according to a given ad-exposure policy. We use the policy-path marginal ROI as a common target estimand to compare and reconcile the Micro and Macro approaches widely used in practice.

One of our contributions (Section 3.1) is to note that the ad-exposure policy (e.g., whether the next impression is allocated to consumers who saw many ads already, or saw just a few) affects the relevant ROI estimate, while most approaches implicitly assume uniform exposure when computing ROIs. We incorporate this important feature into the ensemble fusion solution we develop. A second contribution (Section 4) is to show that under ideal (but likely unrealistic) conditions, such as linear and homogeneous user response, both Micro and Macro yield unbiased estimators of the same underlying policy-path marginal ROI. In realistic settings, however, the estimators deviate in systematically different ways. We characterize this divergence theoretically. Macro and Micro estimates disagree not because one is necessarily correct and the other is necessarily incorrect, but because they lose information in different ways. Macro methods collapse the cross-sectional distribution of exposure histories and user states into aggregate totals, which creates an aggregation bias whenever user response is heterogeneous or marginal lift is nonlinear. Micro methods, in contrast, have other weaknesses: higher variance, sparsity, and specification error in user-level response and policy models. In summary, Macro error is primarily driven by *information loss* while Micro error is primarily driven by *statistical misspecification*. The result is a measurement problem with complementary distortions rather than a simple horse race between two estimators.

Our third contribution uses the fact that Macro and Micro estimates are biased and noisy measurements of the same latent policy-path ROI estimand, meaning that rather than choosing one of them, they should be combined. We introduce an ROI ensemble fusion estimator that combines the Macro and Micro estimates to minimize the mean squared error of the resulting estimate. The framework exploits the asymmetry described above: Macro provides stability but may be structurally biased by aggregation, while Micro is closer to the target estimand but can be noisier and more vulnerable to finite sample effects. By explicitly modeling bias and covariance, the fused estimator places more weight on each estimate when it is informative and discounts it when it is unreliable.

We complement our theoretical analysis with a simulation study (Section 6) and empirical analysis (Section 7). Our extensive simulation study shows that the proposed fusion method more reliably recovers the underlying policy-path ROI than either Macro or Micro approaches alone across environments with different bias–variance configurations, targeting heterogeneity, nonlinear response, and misspecification. We then study a large-scale real-world advertising dataset in which

the latent true ROIs are not observed. Our evaluation criteria in this setting with unknown underlaying true ROI are out-of-sample predictive performance and temporal stability. Our results show that the fused estimator is more stable week over week and yields better predictive performance than relying on either Macro or Micro measurement alone.

Commonly diverging estimates from different ad effectiveness measurement approaches might seem as an issue for managers, but our paper shows the opposite. Since MMM-style and MTA-style procedures can be understood as biased and noisy measurements of a common policy-path marginal ROI estimand, combining them can benefit the decision maker. Our paper provides the charcterization for when this approach might be beneficial, and the tools to achieve a better ad measurement oucome.

2 Related Literature

This paper relates to three literatures. First, a large literature shows that advertising effectiveness is intrinsically difficult to measure, even in settings with rich data or experimental variation. Lewis and Rao [14] show that advertising ROI is often estimated imprecisely even in large-scale experiments. Blake et al. [4] similarly document that conventional non-experimental estimates can substantially overstate paid-search effectiveness. More recent work shows that granular observational data do not, by themselves, resolve the problem: common observational adjustments can perform poorly relative to experiments [9, 10], and even platform-run experiments can be distorted by divergent delivery [5]. What is missing in this literature is not another comparison of estimators in isolation, but a common policy-relevant target that allows the dominant measurement approaches to be interpreted jointly. Our paper contributes that target and uses it to study the macro–micro disagreement as a measurement problem rather than a contest to declare one side correct.

Second, our paper sits between the Macro and Micro measurement traditions. The MMM literature emphasizes aggregate response, dynamics, and managerial relevance for budgeting decisions [12, 8, 13, 17]. The MTA and attribution literature emphasizes user-level paths and richer touchpoint information [15, 1, 3, 16]. At the same time, this literature has also clarified that attribution weights are not themselves the correct object for media budget allocation [6]. Our contribution is

therefore not a new aggregate response specification or a new attribution rule. Instead, we place both paradigms against a common policy-path marginal ROI estimand and show why aggregate and granular procedures diverge from it in structurally different ways.

Third, the paper draws on the estimator-combination literature. A central result in forecasting is that combining imperfect signals can outperform selecting a single forecast, especially when the underlying error structures differ [2, 11]. Our setting is more demanding because Macro and Micro estimates are not merely noisy measurements of the same unbiased quantity; they can also be biased in systematically different directions. We exploit exactly that asymmetry. Macro estimates are often more stable but may suffer from structural aggregation bias, whereas Micro estimates are often closer to the policy-level object but more vulnerable to variance and misspecification. The paper's position is therefore narrow and explicit: we do not argue that Macro should replace Micro, or vice versa. Instead, we contribute a unified policy-path estimand, a theory of why the two measurement systems diverge, and a fusion rule that combines them in a statistically disciplined way.

3 Setup and Notation

In this section, we define a general data-generating process for a single advertising campaign and show how it gives rise to two observational views: a granular user-level view (Micro) and an aggregated campaign-level view (Macro).

Campaign environment. Suppose a marketer runs a campaign over T time periods, indexed by $t \in \{1, \ldots, T\}$, and across C advertising channels, indexed by $c \in \{1, \ldots, C\}$. A channel may correspond to a platform, medium, publisher, or any other delivery environment. Let B_{ct} denote the budget allocated to channel c in period t. Budget is converted into realized impression supply $I_{ct} \in \mathbb{Z}_{\geq 0}$, where the mapping from B_{ct} to I_{ct} may be stochastic because of auctions, inventory conditions, pacing, or other operational frictions.

User state. At each period t, let $\mathcal{A}_t$ denote the set of users eligible to receive advertising. Users are heterogeneous in both baseline outcome propensity and responsiveness to advertising. We represent this heterogeneity through a persistent latent type k_i together with observed user char-

acteristics and state variables. For each user i, let X_{it} denote observed covariates. Let S_{it} denote the user's accumulated exposure history up to time t, generically across channels; canonically, this may be represented through channel-specific adstock, for example $S_{ci,t} = \lambda_c S_{ci,t-1} + z_{cit}$ with $\lambda_c \in [0, 1)$. Let G_{it} denote the user's outcome-history state, such as cumulative conversions, recency, or a decay-weighted stock of past events. We collect the start-of-period information for user i into $H_{it}^- = (X_{it}, S_{i,t-1}, G_{i,t-1}, k_i)$.

Exposure allocation. The ad-delivery policy determines how realized impressions are allocated across users conditional on these states. Because channels may differ in targeting logic, we allow the policy to be channel-specific and denote it by $\pi = \{\pi_{ct}\}$. If impressions in channel c and period t are assigned sequentially, then for the m-th realized impression,

$$\Pr\Big(\text{impression } (c, t, m) \text{ is assigned to user } i \,\Big|\, \mathcal{A}_t, \{H_{jt}^-\}_{j \in \mathcal{A}_t}, \pi_{ct}\Big) = p_{cit}^{(m)}(\pi_{ct}),$$

where $\sum_{i \in \mathcal{A}_t} p_{cit}^{(m)}(\pi_{ct}) = 1$. This formulation accommodates untargeted delivery, retargeting, and other state-dependent allocation rules. Let z_{cit} denote the number of impressions from channel c delivered to user i in period t, so that, by construction, $\sum_{i \in \mathcal{A}_t} z_{cit} = I_{ct}$. Let $Z_{it} = (z_{1it}, \ldots, z_{Cit})$ denote the vector of user-i exposures across channels in period t.

User response and state evolution. Let $Y_{it} \in \mathbb{Z}_{\geq 0}$ denote the outcome generated by user i in period t, interpreted generally as the number of conversion events or other campaign-relevant outcomes. The aggregate campaign outcome is $R_t = \sum_i Y_{it}$. We represent user response by $Y_{it} = f_t(H_{it}^-)$, where $f_t(\cdot)$ is an unknown response function. This notation makes explicit that current outcomes may depend on observed covariates, latent user type, lagged advertising state, and lagged outcome-history state. After exposures are delivered and outcomes are realized, user states evolve in the natural way: advertising state updates through exposure accumulation, and outcome-history state updates through realized outcomes.

Micro and Macro data views. The Micro and Macro datasets are two representations of the same campaign realization. The Micro view retains user-period exposure and outcome histories,

together with the associated state variables:

$$D_{\text{micro}} = \{(i, t, X_{it}, Z_{it}, Y_{it}) : \, i \in \mathcal{A}_t, \; t = 1, \ldots, T\} \, .$$

The Macro view aggregates over users and records only channel-period totals:

$$D_{\text{macro}} = \{(I_{1t}, \ldots, I_{Ct}, R_t)\}_{t=1}^{T} \, .$$

The essential distinction is informational, not ontological: both datasets arise from the same underlying process, but the Macro view discards the cross-sectional joint distribution of user states, exposures, and outcomes. That lost distributional information will be central to the divergence between Macro and Micro measurement developed later in the paper.

3.1 Target Estimand

The object of interest is the marginal effectiveness of advertising under the prevailing delivery policy. In a targeted and dynamic environment, ROI is not a single policy-invariant coefficient. The value of an additional impression depends on who receives it, at what state, and under what allocation rule. For that reason, we define effectiveness relative to the prevailing policy π.

Fix a channel c, a time t, and an evaluation window of length W. Let $N_i(t, t+W] = \sum_{s=t+1}^{t+W} Y_{is}$ denote the number of future events generated by user i over the next W periods. Now consider two potential campaign paths initialized at the same start-of-period states $\{H_{it}^{-}\}_{i \in \mathcal{A}_t}$. In the *baseline* path, the campaign evolves under policy π with no perturbation. In the *perturbed* path, one additional channel-c impression is added at time t and allocated according to the same policy π, after which the campaign continues to evolve under π. We define the *policy-dependent marginal ROI* as the expected incremental future outcome generated by that additional impression:

$$\text{ROI}_c(t; W, \pi) = \mathbb{E}_\pi \left[\sum_{i \in \mathcal{A}_t} \left(N_i^{(+1c)}(t, t+W] - N_i(t, t+W] \right) \,\middle|\, \{H_{it}^{-}\}_{i \in \mathcal{A}_t} \right] . \tag{1}$$

Equation (1) serves as the central estimand of interest in this paper and is expressed in event units: it is the expected number of additional future events generated by one extra impression. It can be viewed as a policy-dependent, finite-horizon analogue of the impulse-response objects used

in dynamic marketing-response analysis [7]. This interpretation is useful because it clarifies that the object is not conceptually foreign to the marketing literature; the key distinction is that here the marginal effect is defined conditional on the prevailing ad-delivery policy. Equation (1) also makes two points explicit. First, advertising effectiveness is policy-dependent because the policy determines the marginal recipient of the additional impression and, through that assignment, the induced distribution of future exposure histories. Second, both Macro and Micro measurement seek to learn the same underlying object, but from different information sets. The remainder of the paper studies what each information set allows the analyst to recover, why the resulting estimates diverge, and how the two signals can be combined.

4 Macro–Micro Divergence

This section characterizes the population targets implicit in Micro and Macro measurement and shows why they generally differ. The source of divergence is informational. Micro data retain the cross-sectional distribution of user states at the moment a marginal impression is allocated, whereas Macro data retain only aggregates. When marginal effectiveness depends on user state, aggregation changes the object that can be recovered. Consequently, Macro disagreement is structural: it can persist even with infinite data and a correctly specified aggregate procedure. By contrast, Micro procedures can in principle target the policy-dependent estimand in Section 3.1, but they are typically more exposed to variance and specification error.

4.1 Population targets under Micro and Macro information

Fix a channel c, an intervention time t, an evaluation window W, and a delivery policy π. Let h denote a generic realization of the user state H_{it}^{-} at time t. Define the state-specific marginal effect

$$\delta_{c,t}^{W}(h) := \mathbb{E}_{\pi}\left[N_i^{(+1c)}(t, t+W] - N_i(t, t+W] \;\middle|\; H_{it}^{-} = h\right].$$

Thus, $\delta_{c,t}^{W}(h)$ is the expected incremental number of future events generated by an additional channel-c impression when the recipient is in state h and the campaign subsequently evolves under policy π.

Let $p_t^\pi(h)$ denote the policy-induced cross-sectional distribution of user states at time t. Since Micro data retain user-level histories, the population Micro target can be written as

$$\mathrm{ROI}_c^{\mathrm{micro},\star}(t; W, \pi) := \int \delta_{c,t}^W(h)\, p_t^\pi(h)\, dh = \mathrm{ROI}_c(t; W, \pi). \tag{2}$$

Equation (2) makes explicit that the Micro target is the state-distribution average of marginal effects under the true policy-induced distribution.

Now let $\mathcal{M}_t$ denote the macro information set at time t, for example the history of aggregate impressions and aggregate outcomes up to time t, possibly augmented by deterministic macro state summaries. Let $R_{t+1:t+W} := \sum_{s=t+1}^{t+W} R_s$ denote aggregate future outcomes over the evaluation window. The oracle Macro target is

$$\mathrm{ROI}_c^{\mathrm{macro},\star}(t; W, \pi) := \mathbb{E}_\pi\left[R_{t+1:t+W}^{(+1c)} - R_{t+1:t+W} \;\middle|\; \mathcal{M}_t\right], \tag{3}$$

where $R_{t+1:t+W}^{(+1c)}$ denotes the aggregate future outcome under the counterfactual path in which one additional channel-c impression is introduced at time t and the campaign then continues under policy π.

The distinction between (2) and (3) is therefore immediate. The Micro target averages state-specific marginal effects under the true cross-sectional distribution that governs marginal effectiveness. The Macro target conditions only on aggregates and therefore does not identify that distribution.

4.2 Structural aggregation bias

Proposition 1 (Structural aggregation bias)**.** *There exists a macro-implied distribution $\widetilde{p}_t(h \mid \mathcal{M}_t)$ over user states, measurable with respect to $\mathcal{M}_t$, such that*

$$\mathrm{ROI}_c^{\mathrm{macro},\star}(t; W, \pi) = \int \delta_{c,t}^W(h)\, \widetilde{p}_t(h \mid \mathcal{M}_t)\, dh. \tag{4}$$

Hence the aggregation bias is

$$
\begin{aligned}
b_{c,t}^{\mathrm{agg}}(W,\pi) &:= \mathrm{ROI}_c^{\mathrm{macro},\star}(t;W,\pi) - \mathrm{ROI}_c(t;W,\pi) \\
&= \int \delta_{c,t}^{W}(h)\Big(\widetilde{p}_t(h \mid \mathcal{M}_t) - p_t^{\pi}(h)\Big)\, dh. \qquad (5)
\end{aligned}
$$

Proposition 1 is the main theoretical result (more on the detailed argument below in Section 4.3). Macro and Micro average the same state-specific marginal effect function $\delta_{c,t}^{W}(h)$, but under different distributions. The Micro target uses the true policy-induced distribution $p_t^{\pi}(h)$. The Macro target uses the macro-implied distribution $\widetilde{p}_t(h \mid \mathcal{M}_t)$ induced by conditioning on aggregates alone. The resulting bias in (5) is structural: it arises from information loss and does not vanish with more data.

The role of repeated events is immediate from the definition of state. Since h includes the outcome-history component $G_{i,t-1}$ in addition to lagged advertising state, repeated-event dynamics enlarge the state space over which marginal effects vary. Aggregation therefore discards not only exposure heterogeneity, but also heterogeneity in outcome history. This strengthens the macro–micro gap whenever future responsiveness depends on prior outcomes as well as prior exposures.

4.3 A second-order bias approximation

Proposition 1 is exact but abstract. A useful approximation is obtained by considering a finite-dimensional state summary $U_t = \psi(h) \in \mathbb{R}^d$, with mean $\bar{U}_t = \mathbb{E}[U_t]$ and covariance matrix $\Sigma_t = \mathrm{Var}(U_t)$. Suppose a macro approximation effectively evaluates marginal effectiveness at the representative state $\bar{U}_t$ rather than integrating over the full cross-sectional distribution. Then the gap between the representative-state approximation and the true cross-sectional average admits a second-order characterization.

Corollary 1 (Curvature–dispersion approximation)**.** *If $\delta_{c,t}^{W}(u)$ is twice continuously differentiable in a neighborhood of $\bar{U}_t$, then*

$$
\delta_{c,t}^{W}(\bar{U}_t) - \mathbb{E}\big[\delta_{c,t}^{W}(U_t)\big] \;\approx\; -\frac{1}{2}\,\mathrm{tr}\Big(\nabla^2 \delta_{c,t}^{W}(\bar{U}_t)\,\Sigma_t\Big). \qquad (6)
$$

A proof is provided in Appendix A. Corollary 1 gives a transparent interpretation of aggregation

bias. The discrepancy grows with two objects: the curvature of the marginal-effect function and the dispersion of user states. If marginal effects are nearly linear in state or if user states are tightly concentrated, the representative-state approximation is accurate. If marginal effects are nonlinear and user states are dispersed, the approximation deteriorates. In targeted campaigns, both sources of distortion are naturally amplified: targeting induces uneven exposure histories, and repeated-event dynamics generate heterogeneity in outcome-history states.

4.4 When does Macro coincide with Micro?

Proposition 2 (Zero-bias benchmark)**.** *If $\delta^W_{c,t}(h) = \delta_0$ for p^π_t-almost every h, then $b^{\text{agg}}_{c,t}(W, \pi) = 0$. A sufficient condition is linear, homogeneous response with no economically relevant state dependence; for example, if $\mathbb{E}[Y_{it} \mid H^-_{it}] = \alpha_t + \sum_{c'=1}^{C} \beta_{c',t} S_{c'i,t-1}$, with coefficients common across users and independent of outcome-history state, then the marginal effect of an additional impression is constant across users and aggregation does not alter the target.*

Proposition 2 is useful precisely because it is restrictive. It describes a benchmark in which Macro and Micro agree because state heterogeneity is irrelevant for marginal effectiveness. Outside that benchmark, Proposition 1 implies that Macro disagreement is primarily structural and informational: aggregates do not identify the state distribution that governs marginal effects. Micro procedures do not suffer from this particular information loss, but they face a different problem. Because they estimate a higher-dimensional object using granular data, they are typically more sensitive to sparsity, variance, and specification error. This asymmetry motivates the fusion approach developed next.

5 Methodology: ROI Ensemble Fusion

This section develops our estimation and fusion procedure for the policy-dependent marginal ROI introduced in Section 3.1. The starting point is the asymmetry established in Section 4: Micro procedures are more closely aligned with the policy-dependent target estimand but are typically more exposed to variance and specification error, whereas Macro procedures are often more stable but can suffer from structural aggregation bias. We treat these procedures as generically correlated, potentially biased measurements of the same latent scalar target and combine them by explicit risk

minimization. We refer to the resulting framework as *ROI Ensemble Fusion* (REF).

5.1 Micro and Macro estimation

For simplicity, in this section we write $\rho \equiv \mathrm{ROI}_c(t; W, \pi)$ for the scalar target associated with a fixed (c, t, W, π).

Micro estimator. The Micro estimator is a plug-in rollout estimator (bootstrap standard errors to account for the plug in nature are described later) built from user-level data. Suppose the user response is modeled parametrically as $f(\cdot; \theta)$. We estimate θ on D_{micro} using a suitable supervised-learning procedure, such as maximum likelihood, least squares, or penalized estimation, and denote the fitted response model by $\widehat{f} = f(\cdot; \widehat{\theta})$. Similarly, let the ad-delivery policy be parameterized by $\pi(\cdot; \tau)$. We estimate τ from the exposure-allocation data in D_{micro}, again using an appropriate supervised-learning procedure, and denote the fitted policy by $\widehat{\pi} = \pi(\cdot; \widehat{\tau})$.

Given $(\widehat{f}, \widehat{\pi})$, the Micro estimator is obtained by forward rollout of the baseline and perturbed paths implied by the fitted models. Specifically,

$$\widehat{\rho}^{\text{micro}} := \mathbb{E}_{\widehat{\pi}}\left[\sum_{i \in \mathcal{A}_t} \left(\widehat{N}_i^{(+1c)}(t, t+W] - \widehat{N}_i(t, t+W]\right) \,\middle|\, \{H_{it}^-\}_{i \in \mathcal{A}_t}\right], \tag{7}$$

where $\widehat{N}_i(t, t+W]$ and $\widehat{N}_i^{(+1c)}(t, t+W]$ denote the fitted future outcome paths under the baseline and perturbed campaigns, respectively, generated using the estimated response model $\widehat{f}$ and the estimated policy $\widehat{\pi}$.

Macro estimator. The Macro estimator is based on aggregate data only. Let $m_t(\mathcal{M}_t) := \mathbb{E}_\pi[R_{t+1:t+W} \mid \mathcal{M}_t]$ denote the aggregate conditional mean over the evaluation window, and let $\widehat{m}_t(\cdot)$ be an estimated macro forecasting model fit on D_{macro} using an appropriate aggregate procedure, such as OLS or penalized regression. The Macro estimator is the finite-difference analogue of the oracle Macro target:

$$\widehat{\rho}^{\text{macro}} := \widehat{m}_t(\mathcal{M}_t^{(+1c)}) - \widehat{m}_t(\mathcal{M}_t), \tag{8}$$

where $\mathcal{M}_t^{(+1c)}$ denotes the counterfactual macro state obtained by adding one channel-c impression at time t.

5.2 Measurement system

Suppose we collect K candidate estimates of the same scalar target in the vector $\widehat{\boldsymbol{\rho}} = (\widehat{\rho}_1, \ldots, \widehat{\rho}_K)^\top$.[1] We represent these estimates by the measurement system

$$\widehat{\boldsymbol{\rho}} = \rho\,\mathbf{1} + b + \varepsilon, \tag{9}$$

where ρ is the latent target, $\mathbf{1}$ is the all-ones vector, $b \in \mathbb{R}^K$ collects systematic biases, and $\varepsilon \in \mathbb{R}^K$ is a zero-mean error with covariance matrix Σ, i.e., $\mathbb{E}[\varepsilon] = 0$ and $\mathrm{Cov}(\varepsilon) = \Sigma$. Correlation across estimators is unrestricted, since all estimators are constructed from the same campaign realization. For Micro estimators, b may reflect response-model misspecification, policy-model misspecification, or rollout approximation error; for Macro estimators, b includes the structural aggregation bias derived in Section 4, along with any reduced-form specification error.

We consider a linear fusion rule of the form

$$\widehat{\rho}^{\,\mathrm{fused}}(w) = w^\top \widehat{\boldsymbol{\rho}}, \qquad \mathbf{1}^\top w = 1. \tag{10}$$

The normalization $\mathbf{1}^\top w = 1$ preserves the target scale. Under (9), $\widehat{\rho}^{\,\mathrm{fused}}(w) - \rho = w^\top b + w^\top \varepsilon$, so the corresponding mean-squared error is

$$\mathrm{MSE}(w) = (w^\top b)^2 + w^\top \Sigma w = w^\top (\Sigma + bb^\top) w. \tag{11}$$

Equation (11) is the population fusion criterion: the optimal weights solve a constrained quadratic program that trades off systematic bias and statistical uncertainty.

The following result provides the generic solution to this class of problems.

Theorem 1 (Quadratic fusion rule)**.** *Let $A \in \mathbb{R}^{K\times K}$ be symmetric and positive definite. Then the*

[1] In the baseline application, $K = 2$ and the entries are the Micro and Macro estimates above, but the framework allows multiple Micro and Macro specifications.

unique solution to

$$\min_{w:\,\mathbf{1}^\top w=1} w^\top A w$$

is

$$w^\star(A) = \frac{A^{-1}\mathbf{1}}{\mathbf{1}^\top A^{-1}\mathbf{1}}, \tag{12}$$

and the minimum value of the criterion is

$$\min_{w:\,\mathbf{1}^\top w=1} w^\top A w = \frac{1}{\mathbf{1}^\top A^{-1}\mathbf{1}}. \tag{13}$$

See Appendix for proof.

In practice, b is unknown and fusion must rely on an estimated bias. Let $\widehat{b} = b+u$ be an estimated bias vector, where $\mathbb{E}[u] = 0$, and let $\Omega := \mathrm{Cov}(u)$ denote the uncertainty of bias estimation. The bias estimate can be correlated with estimator noise; let $\Gamma := \mathrm{Cov}(\varepsilon, u)$ denote this covariance. The corresponding debiased signal vector is

$$\widehat{\boldsymbol{\rho}}^{\text{debiased}} := \widehat{\boldsymbol{\rho}} - \widehat{b} = \rho\,\mathbf{1} + (\varepsilon - u), \tag{14}$$

with effective covariance

$$\Sigma_{\text{eff}} := \mathrm{Cov}(\varepsilon - u) = \Sigma + \Omega - \Gamma - \Gamma^\top. \tag{15}$$

Equations (11) and (15) imply two natural fusion strategies: one that works directly with the original signals, and one that first debiases and then combines them.

5.3 Fusion rules

Approach I: direct fusion. The first strategy fuses the original estimators directly. If b were known, the relevant population criterion would be (11). Since b is estimated in practice, we use the design matrix $A_{\mathrm{I}} := \Sigma + \Omega + bb^\top$, which augments the population criterion by Ω to account for the uncertainty in the estimated bias used to construct the weights. By Theorem 1, if $A_{\mathrm{I}} \succ 0$, then the

optimal direct-fusion weights are

$$w_{\mathrm{I}}^{\star} = \frac{A_{\mathrm{I}}^{-1}\mathbf{1}}{\mathbf{1}^{\top}A_{\mathrm{I}}^{-1}\mathbf{1}}, \tag{16}$$

with minimum criterion value

$$\mathrm{MSE}_{\mathrm{I}}^{\min} = \frac{1}{\mathbf{1}^{\top}A_{\mathrm{I}}^{-1}\mathbf{1}}. \tag{17}$$

Direct fusion does not debias estimator-by-estimator. Instead, it chooses weights by jointly pricing variance, bias magnitude, and bias-estimation uncertainty.

Approach II: debias-first fusion. The second strategy first debiases each estimator using $\widehat{b}$ and then fuses the corrected signals. Since the debiased vector (14) is centered at $\rho\,\mathbf{1}$, the relevant criterion is its residual covariance (15). By Theorem 1, if $\Sigma_{\mathrm{eff}} \succ 0$, then the optimal debias-first weights are

$$w_{\mathrm{II}}^{\star} = \frac{\Sigma_{\mathrm{eff}}^{-1}\mathbf{1}}{\mathbf{1}^{\top}\Sigma_{\mathrm{eff}}^{-1}\mathbf{1}}, \tag{18}$$

with minimum criterion value

$$\mathrm{MSE}_{\mathrm{II}}^{\min} = \frac{1}{\mathbf{1}^{\top}\Sigma_{\mathrm{eff}}^{-1}\mathbf{1}}. \tag{19}$$

Debias-first fusion separates systematic distortion from residual statistical uncertainty. Once the signals are debiased, the fusion problem reduces to a generalized least-squares combination of measurements of the same scalar target.

Under either approach, the optimal fused estimator weakly dominates every individual estimator under the corresponding quadratic criterion. This follows because each individual estimator is itself a feasible point in the constrained optimization problem; see Proposition 4 in Appendix A.1.

Comparison. The two rules solve different optimization problems. Direct fusion prices bias through the criterion, while debias-first fusion removes estimated bias first and then optimizes over residual covariance. Their relative performance is characterized by the following proposition.

Proposition 3 (Comparison of direct and debias-first fusion)**.** *Assume* $A_{\mathrm{I}} \succ 0$ *and* $\Sigma_{\mathrm{eff}} \succ 0$*. Then*

$$\mathrm{MSE}_{\mathrm{II}}^{\min} < \mathrm{MSE}_{\mathrm{I}}^{\min} \quad \Longleftrightarrow \quad \mathbf{1}^\top \Sigma_{\mathrm{eff}}^{-1} \mathbf{1} > \mathbf{1}^\top A_{\mathrm{I}}^{-1} \mathbf{1}. \tag{20}$$

Proof. The claim follows immediately from (17) and (19). □

Thus debias-first fusion is preferable precisely when the gain from removing bias dominates the additional uncertainty induced by bias estimation. This is not guaranteed ex ante; it is an empirical question determined by the relative magnitudes of Σ, Ω, b, and Γ.

5.4 Estimating bias and uncertainty

The fusion rules require estimates of b, Σ, Ω, and Γ, and these objects must be estimated jointly. The framework is agnostic about the specific bias-estimation method. In our application, the Macro bias estimate is informed by Section 4 and may be constructed from a curvature–dispersion approximation, a more general state-reweighting approximation, or a simulation-based calibration. Micro bias may be estimated by cross-fitting, holdout calibration, or simulation under a benchmark micro data-generating process. Denote the resulting bias estimate by $\widehat{b}$.

Because both $\widehat{\rho}^{\mathrm{micro}}$ and $\widehat{b}$ depend on estimated nuisance parameters—including the response-model parameters $\widehat{\theta}$, any latent-state or mixing parameters, and the policy parameters $\widehat{\tau}$—their uncertainty must be propagated into the fusion stage. We do so by re-estimating all nuisance parameters jointly within each resampling draw.

Specifically, we use a joint bootstrap that resamples the underlying Micro observational units, reconstructs the Macro aggregates from each resample, re-estimates all models and nuisance parameters, and recomputes both $\widehat{\boldsymbol{\rho}}$ and $\widehat{b}$. Let $r = 1, \ldots, B$ index bootstrap draws, and let $(\widehat{\boldsymbol{\rho}}^{(r)}, \widehat{b}^{(r)})$ denote the resulting bootstrap replicates. We then compute

$$\widehat{\Sigma} = \mathrm{Cov}_r\left(\widehat{\boldsymbol{\rho}}^{(r)}\right), \qquad \widehat{\Omega} = \mathrm{Cov}_r\left(\widehat{b}^{(r)}\right), \qquad \widehat{\Gamma} = \mathrm{Cov}_r\left(\widehat{\boldsymbol{\rho}}^{(r)}, \widehat{b}^{(r)}\right).$$

These estimates yield

$$\widehat{A}_{\mathrm{I}} = \widehat{\Sigma} + \widehat{\Omega} + \widehat{b}\widehat{b}^\top, \qquad \widehat{\Sigma}_{\mathrm{eff}} = \widehat{\Sigma} + \widehat{\Omega} - \widehat{\Gamma} - \widehat{\Gamma}^\top.$$

If either $\widehat{A}_{\mathrm{I}}$ or $\widehat{\Sigma}_{\mathrm{eff}}$ is ill-conditioned, we regularize by adding a small ridge term before inversion.

5.5 Implementation and implications

Algorithm 1 summarizes the implementation of ROI Ensemble Fusion. The final fused estimator is not fixed ex ante to either approach. Instead, the procedure computes both the direct-fusion and debias-first candidates, evaluates their estimated quadratic risks, and selects the estimator with the lower estimated MSE.

This makes the methodology operational. Rather than forcing the analyst to choose ex ante between Micro and Macro measurement, REF estimates the bias and uncertainty structure of each signal, accounts for their dependence, and selects the lower-risk fusion rule. In this sense, it converts macro–micro disagreement from an informal model-selection problem into a disciplined estimation problem.

A central point is that the procedure does *not* attempt to remove the delivery policy π from the problem. The policy is part of the estimand itself. The object of interest is explicitly policy-dependent, so the role of estimation is to measure that target as accurately as possible, not to partial out the allocation rule and replace it with a policy-invariant quantity. This is why the Micro rollout conditions on $\widehat{\pi}$ and why the fusion step combines alternative measurements of the same policy-dependent target.

The framework has several practical implications. First, the fused estimator is adaptive: when Macro bias is large or its bias estimate is precise, the debias-first rule tends to downweight or more aggressively correct the Macro signal; when the Micro estimate is especially noisy, the procedure can shift weight toward Macro information despite the presence of bias. Second, the framework is not restricted to two estimators. It extends directly to multiple Micro specifications, multiple Macro specifications, or additional external signals, provided each can be represented within (9). In particular, when experimental estimates are available, they can be incorporated as additional low-bias signals within the same fusion system rather than treated as a substitute for it. Third, the same machinery applies whether ROI is measured in event units or monetary units, as long as all candidate estimators target the same scalar estimand after any deterministic scaling.

Finally, the methodology preserves the paper's central distinction: Micro procedures are primarily limited by variance and misspecification, whereas Macro procedures are primarily limited by

Algorithm 1: ROI Ensemble Fusion for $\mathrm{ROI}_c(t; W, \pi)$

Input: channel c, time t, window W, Micro data D_{micro}, Macro data D_{macro}, bootstrap size B.
Fit the user-response model $\widehat{f}$ and delivery-policy model $\widehat{\pi}$ on D_{micro}
compute the Micro estimate $\widehat{\rho}^{\text{micro}}$ by forward rollout.
Fit the aggregate forecasting model $\widehat{m}_t$ on D_{macro}; compute the Macro estimate $\widehat{\rho}^{\text{macro}}$ by finite difference.
Form $\widehat{\boldsymbol{\rho}} = (\widehat{\rho}^{\text{micro}}, \widehat{\rho}^{\text{macro}})^\top$ and construct the bias estimate $\widehat{b}$.
for $r = 1, \ldots, B$ **do**
    Resample the underlying Micro observational units with replacement.
    Reconstruct the Macro aggregates from the resampled Micro data.
    Re-estimate all nuisance parameters and models, including response, latent-state, and policy parameters.
    Recompute $\widehat{\boldsymbol{\rho}}^{(r)}$ and $\widehat{b}^{(r)}$.
end for
Compute $\widehat{\Sigma}$, $\widehat{\Omega}$, and $\widehat{\Gamma}$ from the bootstrap replicates.
Form $\widehat{A}_{\mathrm{I}} = \widehat{\Sigma} + \widehat{\Omega} + \widehat{b}\,\widehat{b}^\top$ and $\widehat{\Sigma}_{\text{eff}} = \widehat{\Sigma} + \widehat{\Omega} - \widehat{\Gamma} - \widehat{\Gamma}^\top$.
Compute the candidate weights $\widehat{w}_{\mathrm{I}}^\star = \frac{\widehat{A}_{\mathrm{I}}^{-1}\mathbf{1}}{\mathbf{1}^\top \widehat{A}_{\mathrm{I}}^{-1}\mathbf{1}}$, and $\widehat{w}_{\mathrm{II}}^\star = \frac{\widehat{\Sigma}_{\text{eff}}^{-1}\mathbf{1}}{\mathbf{1}^\top \widehat{\Sigma}_{\text{eff}}^{-1}\mathbf{1}}$.
Compute the two candidate fused estimates $\widehat{\rho}_{\mathrm{I}}^{\text{fused}} = \widehat{w}_{\mathrm{I}}^{\star\top}\widehat{\boldsymbol{\rho}}$, and $\widehat{\rho}_{\mathrm{II}}^{\text{fused}} = \widehat{w}_{\mathrm{II}}^{\star\top}(\widehat{\boldsymbol{\rho}} - \widehat{b})$.
Compute the corresponding estimated risks $\mathrm{MSE}_{\mathrm{I}}^{\min} = \frac{1}{\mathbf{1}^\top \widehat{A}_{\mathrm{I}}^{-1}\mathbf{1}}$, and $\mathrm{MSE}_{\mathrm{II}}^{\min} = \frac{1}{\mathbf{1}^\top \widehat{\Sigma}_{\text{eff}}^{-1}\mathbf{1}}$.
Select the estimator with the lower estimated MSE.
Output: fused estimate $\widehat{\rho}^{\text{fused}}$, selected fusion rule, optimal weights, and estimated risk.

structural aggregation bias. Fusion is valuable precisely because these limitations differ. It exploits the fact that one estimator may be biased but stable while another is less biased but noisier, and turns that tension into an explicit risk-minimization problem.

6 Simulation

This section evaluates ROI Ensemble Fusion in a controlled environment where the full campaign data-generating process is known. The simulation is designed to mirror the measurement problem studied in Sections 3–5: a single dynamic campaign generates one underlying realization, and that same realization is observed through both a user-level view and an aggregate view. This allows us to compare Micro, Macro, and fused estimators against the same oracle policy-path target and to study directly the complementary failure modes emphasized by the theory: Macro is vulnerable to structural aggregation bias, whereas Micro is typically closer to the target but more sensitive to variance and misspecification.

The main purpose of the simulation is therefore *measurement evaluation*. We vary the underlying features of the campaign environment that should change the relative bias and variance of Micro and Macro estimation, and we ask whether the proposed fusion rule adapts in the correct direction. A downstream budget-allocation exercise is useful as an implication of improved measurement, but

it is not the primary focus of this section. Throughout, the target object is the policy-dependent windowed marginal ROI $\rho_c(t; W, \pi) = \mathrm{ROI}_c(t; W, \pi)$ defined in Section 3.1.

6.1 Simulation design and baseline setup

We begin from a baseline configuration chosen to be realistic, non-degenerate, and diagnostically informative. The campaign horizon is $T = 60$, the number of channels is $C = 3$, the user population size is $N = 600$, the forward evaluation window is $W = 7$, and the latent heterogeneity structure has $K = 2$ classes with mixing weights $\omega = (0.4, 0.6)$. This baseline is deliberately balanced: it is rich enough to generate meaningful Micro–Macro disagreement, but not so extreme that one estimator becomes trivially dominated everywhere.

The simulation is built around grouped daily user-level data. Day-t outcomes are generated from the start-of-day state, while day-t exposures affect future outcomes only through next-period state transitions. This timing convention matches the empirical structure motivating the paper and isolates the paper's central issue—measurement under dynamic policy-driven delivery—without imposing an unobserved within-day event ordering.

Channel-day budgets are drawn independently from channel-specific ranges,

$$B_{ct} \sim \mathrm{Unif}(B_{c,\min}, B_{c,\max}),$$

with $(B_{c,\min}, B_{c,\max}) \in \{(150, 200), (250, 350), (200, 300)\}$ and cost-per-impression vector $\kappa = (0.50, 0.25, 0.10)$. Realized impressions are then

$$I_{ct} = \left\lfloor \frac{B_{ct}}{\kappa_c} \right\rfloor.$$

This design produces three channels with distinct exposure intensities while keeping the spend-to-impression mapping transparent.

Each user carries two dynamic state variables. The first is channel-specific adstock,

$$S_{ci,t} = \lambda_{c,k_i} S_{ci,t-1} + z_{cit},$$

with class-specific decay profiles

$$(\lambda_{c,1})_{c=1}^{3} = (0.45, 0.30, 0.20), \qquad (\lambda_{c,2})_{c=1}^{3} = (0.75, 0.55, 0.40).$$

Thus one latent class is both more persistent and more responsive to advertising, which is important for generating policy-induced state concentration. The second state variable is outcome-history stock,

$$G_{it} = \delta G_{i,t-1} + Y_{i,t-1}, \qquad \delta = 0.30,$$

which captures repeated-event dependence in a parsimonious way.

Exposure assignment follows a targeted softmax policy. Within each (c, t), impressions are allocated sequentially with probabilities proportional to

$$\exp\{\tau \log(1 + S_{ci,t}^{(m-1)})\},$$

where $\tau \geq 0$ governs targeting intensity. When $\tau = 0$, delivery is approximately uniform at random; as τ increases, delivery becomes increasingly concentrated on users with high current adstock. This mechanism is central to the simulation design: it creates exactly the type of policy-induced selection that the Macro estimator cannot observe once user-level state heterogeneity has been aggregated away. Unless otherwise stated, the baseline uses $\tau = 0.8$, which represents moderate but nontrivial targeting.

User outcomes are generated from a two-part count process. Let $D_{it} = \mathbf{1}\{Y_{it} \geq 1\}$ denote whether user i has any event on day t. Conditional on the start-of-day state and latent class,

$$D_{it} \sim \text{Bernoulli}(q_{it}), \qquad \text{logit}(q_{it}) = \alpha_{k_i} + \sum_{c=1}^{C} \beta_{c,k_i} S_{ci,t-1} + \gamma_{k_i} G_{it} + \eta^\top X_t + \sum_{c<d} \beta_{cd,k_i}^{\text{int}} S_{ci,t-1} S_{di,t-1},$$

and, conditional on $D_{it} = 1$,

$$Y_{it} \mid \{D_{it} = 1\} \sim \text{ZTP}(\mu_{it}), \qquad \log \mu_{it} = \bar{\alpha}_{k_i} + \sum_{c=1}^{C} \bar{\beta}_{c,k_i} S_{ci,t-1} + \bar{\gamma}_{k_i} G_{it} + \bar{\eta}^\top X_t,$$

where ZTP denotes the zero-truncated Poisson distribution. The baseline incidence intercepts are

$\alpha = (-3.5, -2.0)$, so outcomes are sparse but not pathologically rare. The response coefficients are chosen so that the high-persistence class is also more ad-responsive, which creates a clean mechanism for Macro bias under targeted delivery. At the same time, the positive-count stage is deliberately weaker than the incidence stage, so most of the advertising effect operates through whether an event occurs rather than through unrealistically large conditional counts. The resulting panel contains many zeros, occasional repeated events, meaningful user heterogeneity, and substantial variation in local marginal ROI.

Several features of the baseline are chosen specifically to make the measurement comparison informative. First, the campaign is long enough for adstock to reach its steady regime before the main evaluation points, so comparisons are not driven by a transient burn-in artifact. Second, $N = 600$ is large enough for the Micro estimator to be feasible, but not so large that its variance vanishes and fusion becomes mechanically identical to Micro. Third, the Macro model is always given the correct population-weighted adstock decay, so when Macro fails it fails for the reason emphasized by the theory—aggregation and selection bias—rather than because it was handicapped by an obviously incorrect carryover specification.

6.2 Oracle target, implementation, and evaluation protocol

For each campaign realization and each evaluation pair (c, t), the oracle target is computed directly from the true DGP by forward rollout. Starting from the realized states at time t, we simulate the baseline campaign path under the true policy and the perturbed path that injects one additional channel-c impression at time t, again allocated according to the same policy. The oracle $\rho_c(t; W, \pi)$ is the difference in future window outcomes averaged over users and rollout draws. In the implementation, oracle rollouts use common random numbers and 500 inner draws per evaluation point to reduce Monte Carlo noise.

The simulated data are then treated as observed data and the estimators are fit exactly as in the proposed methodology. The Micro estimator fits the user-level response model together with a delivery-policy model and evaluates ROI by forward rollout under the fitted objects. The Macro estimator fits a dynamic aggregate count model and evaluates ROI by finite-difference rollout at the aggregate level. The fused estimator combines these measurements using the procedure in Section 5. Full estimation and computation details are deferred to Appendix A.2 to keep the main

simulation narrative focused on measurement behavior rather than algorithmic bookkeeping.

We run $M = 200$ independent Monte Carlo replications. In each replication, we simulate a full campaign, compute the oracle ROI, fit Micro and Macro, and then apply the fusion rule. Evaluation is carried out at four intervention times,

$$t \in \{15, 30, 45, 53\},$$

which span the campaign after the initial adstock burn-in and include a late-horizon case near the sample boundary. For fusion, second-moment objects are estimated using a joint bootstrap with $B = 300$ user-level resamples. In each bootstrap draw, users are resampled, macro aggregates are reconstructed from the resampled micro panel, and all nuisance objects are re-estimated. This preserves the dependence structure between Micro and Macro measurements, which is essential for valid fusion weights.

Our primary metric is mean-squared error relative to the oracle target. For estimator v, we report

$$\mathrm{Bias}_v = \frac{1}{M}\sum_{m=1}^{M}\big(\widehat{\rho}_v^{(m)} - \rho^{(m)}\big), \qquad \mathrm{Var}_v = \mathrm{Var}_m\big(\widehat{\rho}_v^{(m)}\big), \qquad \mathrm{MSE}_v = \frac{1}{M}\sum_{m=1}^{M}\big(\widehat{\rho}_v^{(m)} - \rho^{(m)}\big)^2.$$

We also report average fusion weights and empirical Micro–Macro correlations, since these quantities help explain when fusion gains are large or small.

6.3 Main performance results

We begin with the baseline design and compare the proposed fused estimator against four benchmarks: Micro alone, Macro alone, the naive equal-weight average, and variance-only weighting. This baseline comparison serves two purposes. First, it establishes the central empirical fact motivating the method: Micro and Macro fail differently even when they are measuring the same latent target. Second, it provides the reference point against which the mechanism experiments below should be interpreted.

Table **??** reports the main measurement results across channels and evaluation times. The expected pattern is a clear bias–variance contrast. Micro is typically closer to the oracle target

but more dispersed across Monte Carlo replications; Macro is typically tighter but systematically displaced when targeting and heterogeneity induce nontrivial aggregation bias. The fused estimator should improve on both by exploiting precisely this complementarity. Figure **??** visualizes the same comparison through the full sampling distribution of the estimators, which is useful because it shows directly that fusion is not merely shifting means, but reducing economically relevant dispersion around the oracle target.

We also report parameter recovery for the user-level model in Appendix Table **??**. That table is not meant to be a separate contribution; its role is diagnostic. It verifies that the Micro estimator is recovering the main latent structure of the DGP in the baseline design and confirms that the main simulation findings are not driven by pathological estimation failure. The key object of interest in the paper remains ROI measurement, not parameter recovery per se, but the latter is useful for reassuring the reader that the baseline calibration is well behaved.

6.4 Mechanism experiments

The remaining experiments are organized around the bias and variance channels highlighted by the theory. In each case, we vary one feature of the DGP that should alter the relative reliability of Micro and Macro estimation, and we test whether the fusion rule responds appropriately.

6.4.1 Impact of sample size

We first study sample size because it primarily changes the statistical precision of the Micro estimator. Specifically, we vary

$$N \in \{75, 150, 300, 600, 1200\},$$

while scaling budgets proportionally with N so that per-user impression intensity remains comparable across designs. This is important: without budget scaling, changing N would also change the steady-state adstock regime and therefore confound the interpretation of the experiment.

This experiment isolates the variance side of the problem. Macro bias is structural and should not disappear with larger N, whereas Micro variance should fall with sample size. Accordingly, the key question is not simply whether Micro eventually overtakes Macro, but whether the crossover occurs where the theory suggests and whether fusion remains near-best on both sides of that

crossover. In small samples, Macro may be preferable despite its bias because Micro is too noisy; in large samples, Micro should dominate as variance falls. Fusion should interpolate smoothly between these regimes rather than forcing an ex ante choice.

6.4.2 Impact of outcome sparsity

We next study outcome sparsity because it is a primary determinant of Micro estimation difficulty in digital advertising applications. To do so, we vary the incidence intercepts across four regimes,

$$[-5.5, -4.0], \quad [-4.0, -2.5], \quad [-3.5, -2.0], \quad [-2.5, -1.0],$$

corresponding to very sparse, sparse, moderate, and dense outcome environments.

The purpose of this experiment is to isolate a distinct variance mechanism. When outcomes are rare, the Micro model still has access to the correct user-level state variables, but the positive-count component is estimated from very few event days. This should raise Micro variance sharply even when structural bias remains limited. Macro, by contrast, does not confront the same event-level sparsity problem because it works with aggregate counts. The relevant question is therefore whether fusion recognizes sparse-outcome regimes as cases in which the Macro signal is useful mainly as a stabilizing device rather than as a low-bias estimator.

6.4.3 Impact of heterogeneity and targeting

We then turn to the main bias mechanism on the Macro side. The theory predicts that Macro disagreement should become more severe when the policy induces stronger concentration in user states and when the cross-sectional distribution of user response becomes more dispersed. We examine these channels in two ways.

First, we vary latent heterogeneity by changing the spread in class-specific adstock decays while holding the population-weighted average decay fixed. The spread values are

$$\{0.10, 0.30, 0.50, 0.70\}.$$

Holding the average fixed is important because it keeps the Macro carryover specification correctly

centered. As a result, any deterioration in Macro performance can be attributed to aggregation of heterogeneous user states rather than to a trivial decay misspecification.

Second, we vary targeting intensity over

$$\tau \in \{0, 0.4, 0.8, 1.5, 2.5, 4.0\}.$$

The case $\tau = 0$ provides the random-delivery benchmark. As τ increases, impressions become increasingly concentrated on high-adstock users, so the aggregate state becomes less representative of the policy-path distribution that governs true marginal ROI. This is exactly the selection channel that Macro cannot recover after aggregation. Accordingly, this experiment is designed to assess how quickly Macro bias grows with targeting intensity and whether the fusion rule correspondingly shifts weight toward Micro.

6.4.4 Impact of misspecification

Finally, we study robustness to Micro misspecification. In the baseline design, Micro is intended to be the lower-bias but higher-variance signal. That role becomes less secure if the user-level model is misspecified. We therefore refit the simulated data under three alternative Micro specifications: $K_{\text{fit}} = 1$, $K_{\text{fit}} = 3$, and a wrong-link variant. These cases capture underfitting of latent heterogeneity, overfitting-induced variance inflation, and functional-form error, respectively.

This experiment asks whether fusion remains useful when Micro is no longer a clean anchor. If the methodology is behaving correctly, it should not mechanically favor Micro merely because Micro is more granular. Rather, it should recognize when Micro has become unstable or biased and reallocate weight accordingly. This is the strongest test of whether the method is genuinely adaptive rather than simply a sophisticated justification for preferring one measurement scale.

6.5 What the simulation is designed to show

Taken together, the simulation section is designed to answer three questions. First, does Macro exhibit the structural bias predicted by Section 4 when targeting and heterogeneity become more pronounced? Second, does Micro exhibit the complementary failure mode predicted by the theory—typically lower structural bias, but higher variance and greater sensitivity to sparsity and misspec-

ification? Third, does the fusion rule adapt in the correct direction as these conditions change?

The baseline comparison establishes the central Micro–Macro tradeoff. The sample-size and sparsity experiments isolate the main variance channels affecting Micro. The heterogeneity and targeting experiments isolate the main structural-bias channels affecting Macro. The misspecification experiment tests whether fusion remains useful once Micro itself becomes imperfect. The point of the simulation is therefore not merely to show that fusion improves average MSE in one calibration, but to demonstrate that it improves for the reasons predicted by the theory.

7 Empirical Application

We apply the proposed framework to a large-scale dataset from a major advertiser on Google, encompassing millions of daily impressions across multiple channels. In contrast to the simulation setting, the empirical application does not reveal the true policy-path ROI or the oracle value function. Accordingly, our empirical goal is not to recover a known ground truth, but rather to evaluate whether the proposed fusion framework delivers a more accurate and more stable measurement signal than standalone Micro or Macro approaches.

We therefore adopt a rolling out-of-sample predictive validation strategy. Specifically, we compare the Micro, Macro, and Fused estimators in terms of their ability to predict future aggregate outcomes over holdout periods. This predictive exercise is directly relevant to the paper's decision layer: a measurement procedure that more accurately captures the underlying response surface should yield better out-of-sample forecasts of future performance under observed spend patterns. We show that the Fused estimator achieves superior predictive accuracy, as measured by lower out-of-sample MAPE, relative to standalone Macro (MMM-style) and Micro (MTA-style) procedures.

Because true empirical ROI is unobserved, we complement predictive validation with a second criterion: the stability of the resulting decision recommendations. Using the rollout-based allocation framework developed earlier, we compare the implied budget recommendations generated by the Micro, Macro, and Fused estimators over time. We show that the Fused estimator yields materially more stable recommendations, exhibiting lower week-over-week volatility and narrower uncertainty in the implied allocation shares. From a managerial perspective, this stability is important: even when oracle regret cannot be measured directly, a method that combines stronger predictive performance with more robust and less erratic budget recommendations provides a more credible basis for strategic resource allocation.

Taken together, the empirical analysis complements the simulation results. The simulation demonstrates that fusion improves measurement and downstream decisions when the truth is known; the empirical application shows that, in a realistic observational setting where truth is unobserved, the same framework delivers better out-of-sample performance and more stable allocation guidance.

8 Conclusion

In this section, we summarize our findings and discuss their broader implications for marketing practice. We suggest that firms should abandon the binary choice between MMM and MTA in favor of a unified approach.

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Acknowledgments

A Proof of Corollary 1

For notational simplicity, write $\delta(u) \equiv \delta^W_{c,t}(u)$, $\bar{U} \equiv \bar{U}_t$, and $\Sigma \equiv \Sigma_t$. Since $\delta(\cdot)$ is twice continuously differentiable in a neighborhood of $\bar{U}$, a second-order Taylor expansion of $\delta(U_t)$ around $\bar{U}$ yields

$$\delta(U_t) = \delta(\bar{U}) + \nabla\delta(\bar{U})^\top (U_t - \bar{U}) + \frac{1}{2}(U_t - \bar{U})^\top \nabla^2\delta(\bar{U})(U_t - \bar{U}) + r(U_t), \tag{21}$$

where the remainder satisfies

$$\frac{r(U_t)}{\|U_t - \bar{U}\|^2} \to 0 \qquad \text{as } U_t \to \bar{U}. \tag{22}$$

Taking expectations of (21) gives

$$\mathbb{E}[\delta(U_t)] = \delta(\bar{U}) + \nabla\delta(\bar{U})^\top \mathbb{E}[U_t - \bar{U}] + \frac{1}{2}\,\mathbb{E}\Big[(U_t - \bar{U})^\top \nabla^2\delta(\bar{U})(U_t - \bar{U})\Big] + \mathbb{E}[r(U_t)]. \tag{23}$$

Since $\bar{U} = \mathbb{E}[U_t]$, the first-order term vanishes:

$$\mathbb{E}[U_t - \bar{U}] = 0. \tag{24}$$

Using the standard identity $\mathbb{E}[X^\top A X] = \operatorname{tr}(A \operatorname{Var}(X))$ for any mean-zero random vector X and constant matrix A, the quadratic term becomes

$$\mathbb{E}\Big[(U_t - \bar{U})^\top \nabla^2\delta(\bar{U})(U_t - \bar{U})\Big] = \operatorname{tr}\Big(\nabla^2\delta(\bar{U})\,\Sigma\Big). \tag{25}$$

Substituting (24) and (25) into (23) yields

$$\mathbb{E}[\delta(U_t)] = \delta(\bar{U}) + \frac{1}{2}\operatorname{tr}\Big(\nabla^2\delta(\bar{U})\,\Sigma\Big) + \mathbb{E}[r(U_t)]. \tag{26}$$

Rearranging gives the exact identity

$$\delta(\bar{U}) - \mathbb{E}[\delta(U_t)] = -\frac{1}{2}\operatorname{tr}\Big(\nabla^2\delta(\bar{U})\,\Sigma\Big) - \mathbb{E}[r(U_t)]. \tag{27}$$

Therefore, whenever the remainder term $\mathbb{E}[r(U_t)]$ is negligible relative to the second-order

term—for example, when the distribution of U_t is sufficiently concentrated around $\bar{U}$ or higher-order terms are small—we obtain the approximation

$$\delta(\bar{U}) - \mathbb{E}[\delta(U_t)] \;\approx\; -\frac{1}{2}\,\mathrm{tr}\Big(\nabla^2\delta(\bar{U})\,\Sigma\Big), \tag{28}$$

which is exactly (6). □

A.1 Additional proofs for the fusion framework

Proof of Theorem 1. Consider the constrained quadratic program

$$\min_{w:\,\mathbf{1}^\top w=1} w^\top A w,$$

where A is symmetric and positive definite. The Lagrangian is

$$\mathcal{L}(w,\lambda) = w^\top A w + \lambda(\mathbf{1}^\top w - 1).$$

The first-order condition with respect to w is

$$\nabla_w \mathcal{L}(w,\lambda) = 2Aw + \lambda\mathbf{1} = 0,$$

which implies

$$w = -\frac{\lambda}{2}A^{-1}\mathbf{1}.$$

Imposing the constraint $\mathbf{1}^\top w = 1$ gives

$$-\frac{\lambda}{2}\mathbf{1}^\top A^{-1}\mathbf{1} = 1, \qquad \text{so} \qquad \lambda = -\frac{2}{\mathbf{1}^\top A^{-1}\mathbf{1}}.$$

Substituting back yields

$$w^\star(A) = \frac{A^{-1}\mathbf{1}}{\mathbf{1}^\top A^{-1}\mathbf{1}},$$

which is (12). Since $A \succ 0$, the objective is strictly convex on the feasible set, so the optimizer is unique.

To obtain the minimum value, substitute $w^\star(A)$ into the objective:

$$w^\star(A)^\top A w^\star(A) = \frac{\mathbf{1}^\top A^{-1} A A^{-1} \mathbf{1}}{(\mathbf{1}^\top A^{-1}\mathbf{1})^2} = \frac{\mathbf{1}^\top A^{-1}\mathbf{1}}{(\mathbf{1}^\top A^{-1}\mathbf{1})^2} = \frac{1}{\mathbf{1}^\top A^{-1}\mathbf{1}},$$

which is (13). □

Proposition 4 (Fusion weakly dominates each individual estimator)**.** *Suppose* $A_{\mathrm{I}} \succ 0$ *and* $\Sigma_{\mathrm{eff}} \succ 0$*. Then:*

1. *Under Approach I,*

$$\mathrm{MSE}_{\mathrm{I}}^{\min} \le e_k^\top A_{\mathrm{I}} e_k \qquad \textit{for every } k \in \{1, \dots, K\}, \tag{29}$$

   where e_k *is the* k*th coordinate basis vector.*

2. *Under Approach II,*

$$\mathrm{MSE}_{\mathrm{II}}^{\min} \le e_k^\top \Sigma_{\mathrm{eff}} e_k \qquad \textit{for every } k \in \{1, \dots, K\}. \tag{30}$$

Equality holds only if the corresponding individual estimator is itself optimal under the relevant quadratic criterion.

Proof. For any $k \in \{1, \dots, K\}$, the coordinate basis vector e_k satisfies $\mathbf{1}^\top e_k = 1$ and is therefore feasible for both constrained optimization problems. By Theorem 1, Approach I minimizes $w^\top A_{\mathrm{I}} w$ over the feasible set, so

$$\mathrm{MSE}_{\mathrm{I}}^{\min} = \min_{w:\, \mathbf{1}^\top w = 1} w^\top A_{\mathrm{I}} w \le e_k^\top A_{\mathrm{I}} e_k$$

for every k, which proves (29). The same argument applied to the criterion $w^\top \Sigma_{\mathrm{eff}} w$ yields (30). Equality can hold only if the corresponding basis vector itself solves the relevant optimization problem. □

A.2 Simulation estimation and implementation details

This appendix collects the estimation and computation details used in the simulation. These details are referenced in Section 6 but separated from the main text so that the simulation section can

focus on measurement behavior rather than algorithmic specification.

Oracle computation. For each realization and evaluation pair (c, t), the oracle target $\rho_c(t; W, \pi)$ is computed by forward rollout under the true DGP. Starting from the realized user states at time t, we simulate the baseline path and the perturbed path that injects one additional channel-c impression at time t, with both paths evolving thereafter under the same policy π. Oracle rollout uses common random numbers and $B_{\text{rollout}} = 500$ inner draws.

Micro estimator. The analyst observes grouped daily micro data

$$D_{\text{micro}} = \big\{(i, t, z_{1it}, \ldots, z_{Cit}, Y_{it}, D_{it})\big\}, \qquad D_{it} = \mathbf{1}\{Y_{it} \geq 1\}.$$

From these data, the analyst reconstructs adstock and event-history states,

$$\widehat{S}_{ci,t} = \lambda_c \widehat{S}_{ci,t-1} + z_{cit}, \qquad \widehat{G}_{it} = \delta \widehat{G}_{i,t-1} + Y_{i,t-1}.$$

The baseline Micro model is a two-part latent-class specification. Conditional on class k,

$$D_{it} \mid H_{it}^{-}, k \sim \text{Bernoulli}(q_{it}(k)), \qquad \text{logit}(q_{it}(k)) = \alpha_k + \sum_{c=1}^{C} \beta_{c,k} \widehat{S}_{ci,t-1} + \gamma_k \widehat{G}_{it} + \eta^\top X_t + \sum_{c<d} \beta_{cd,k} \widehat{S}_{ci,t-1} \widehat{S}_{di,t-1},$$

and, conditional on $D_{it} = 1$,

$$Y_{it} \mid \{D_{it} = 1\}, H_{it}^{-}, k \sim \text{ZTP}(\mu_{it}(k)), \qquad \log \mu_{it}(k) = \bar{\alpha}_k + \sum_{c=1}^{C} \bar{\beta}_{c,k} \widehat{S}_{ci,t-1} + \bar{\gamma}_k \widehat{G}_{it} + \bar{\eta}^\top X_t.$$

Latent classes are estimated by EM. Writing Θ_k for class-specific parameters and ω_k for class weights, the user-level mixture likelihood is

$$L_i(\Theta) = \sum_{k=1}^{K} \omega_k L_i(\Theta_k).$$

The E-step updates posterior class probabilities, and the M-step updates (ω_k, Θ_k) by weighted maximization. The fitted Micro ROI is then computed by rollout under the fitted response model

and fitted policy model:

$$\widehat{\rho}_c^{\text{micro}}(t;W) = \frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K}\widehat{\gamma}_{ik}\,\widehat{\mathbb{E}}_{\widehat{\pi}}\Big[N_i^{(+1_c)}(t,t+W] - N_i(t,t+W]\,\Big|\,H_{it}^{-},k\Big].$$

Macro estimator. The analyst also observes the aggregate time series

$$D_{\text{macro}} = \big\{(I_{1t},\ldots,I_{Ct},R_t)\big\}_{t=1}^{T}, \qquad R_t = \sum_{i=1}^{N} Y_{it}.$$

Aggregate adstock is constructed as

$$A_{c,t} = \bar{\lambda}_c A_{c,t-1} + I_{ct},$$

where $\bar{\lambda}_c$ is the population-weighted decay used throughout the simulation. The baseline Macro model is a dynamic aggregate count specification,

$$R_t \mid \mathcal{M}_{t-1} \sim \text{Poisson}(\Lambda_t), \qquad \log\Lambda_t = \theta_0 + \sum_{c=1}^{C}\theta_c A_{c,t} + \sum_{c<d}\theta_{cd}A_{c,t}A_{d,t} + \varphi R_{t-1} + \zeta^{\top}X_t.$$

The fitted Macro ROI is evaluated by finite-difference rollout through the aggregate model:

$$\widehat{\rho}_c^{\text{macro}}(t;W) = \widehat{M}_t(\mathcal{M}_t^{(+1_c)}) - \widehat{M}_t(\mathcal{M}_t),$$

where $\widehat{M}_t(\cdot)$ denotes the fitted predictor of aggregate window outcomes over $(t,t+W]$.

Fusion implementation. Given the original-data estimator vector

$$\widehat{\boldsymbol{\rho}} = \big(\widehat{\rho}^{\text{micro}},\widehat{\rho}^{\text{macro}}\big)^{\top},$$

fusion requires joint estimation of the Micro and Macro uncertainty objects. We use a user-level bootstrap with $B = 300$ replications. In bootstrap draw r, users are resampled with replacement, macro aggregates are reconstructed from the resampled micro panel, all nuisance models are refit,

and both $\widehat{\boldsymbol{\rho}}^{(r)}$ and the corresponding bias estimate $\widehat{b}^{(r)}$ are recomputed. This yields

$$\widehat{\Sigma} = \mathrm{Cov}_r\big(\widehat{\boldsymbol{\rho}}^{(r)}\big), \qquad \widehat{\Omega} = \mathrm{Cov}_r\big(\widehat{b}^{(r)}\big), \qquad \widehat{C} = \mathrm{Cov}_r\big(\widehat{\boldsymbol{\rho}}^{(r)}, \widehat{b}^{(r)}\big).$$

These matrices are then used exactly as described in Section 5 to construct the direct-fusion and debias-first candidates and select the lower-risk rule.

Monte Carlo protocol. Each reported simulation result is based on $M = 200$ independent campaign replications. In each replication we: (i) simulate one full campaign realization; (ii) compute the oracle target; (iii) fit the Micro and Macro estimators; (iv) apply fusion; and (v) record estimator bias, variance, MSE, and the selected fusion weights. The main evaluation times are

$$t \in \{15, 30, 45, 53\},$$

which span the campaign after the initial burn-in and include a late-horizon case near the sample boundary.

Reported metrics. For each estimator $v \in \{\mathrm{micro}, \mathrm{macro}, \mathrm{fusion}\}$, we report

$$\mathrm{Bias}_v = \frac{1}{M}\sum_{m=1}^{M}\big(\widehat{\rho}_v^{(m)} - \rho^{(m)}\big), \qquad \mathrm{Var}_v = \mathrm{Var}_m\big(\widehat{\rho}_v^{(m)}\big), \qquad \mathrm{MSE}_v = \frac{1}{M}\sum_{m=1}^{M}\big(\widehat{\rho}_v^{(m)} - \rho^{(m)}\big)^2.$$

We also report average fusion weights and empirical Micro–Macro correlations, since these are useful for interpreting when the gains from fusion are driven mainly by bias correction, variance reduction, or both.

Secondary decision layer. A simple downstream budget-allocation exercise is also considered as a secondary implication of improved measurement. Because the paper's main contribution is the measurement layer, these decision results are reported separately and interpreted as downstream consequences of better ROI estimation rather than as the central object of the simulation.